concluding the proof of the existence of a recovery sequence for a smooth y.

To conclude the proof, we first observe that E_j^{K} is continuous with respect to the $W^{2,2}$ topology. Let

$$W^{2,2}_{\mathrm{iso},j}(S;\mathbb{R}^3) := \{y \in W^{2,2}(S;\mathbb{R}^3) \,:\, \nabla y^T \nabla y = (U_j^2)'\}\,.$$

For every $y \in W^{2,2}_{\mathrm{iso},j}(S;\mathbb{R}^3)$, arguing as in (38), we have $(\nabla y)^T \nabla \nu = -\nabla^2 y \nu$, where ν is defined as in Lemma 3.6. Given a sequence $(y_n) \subset W^{2,2}_{\mathrm{iso},j}(S;\mathbb{R}^3)$ such that $y_n \to y$ in $W^{2,2}(S;\mathbb{R}^3)$ we have, up to a subsequence, $\nabla^2 y_n \to \nabla^2 y$ almost everywhere in S. Let ν_n and ν be defined as in Lemma 3.6 for y_n and y, respectively. Then, $\nu_n \to \nu$ in $L^1(S;\mathbb{R}^3)$, thus, up to subsequences, $\nu_n \to \nu$ almost everywhere in S. Hence, $\nabla^2 y_n \nu_n \to \nabla^2 y \nu$ and by dominated convergence $E_j^{\mathrm{K}}(y_n) \to E_j^{\mathrm{K}}(y)$. Moreover, the set of functions $y \in C^\infty(\bar{S};\mathbb{R}^3)$ satisfying $(\nabla y)^T \nabla y = (U_j^2)'$ is dense in $W^{2,2}_{\mathrm{iso},j}(S;\mathbb{R}^3)$. Indeed, let $\varepsilon > 0$ and pick an isometric embedding $y \in W^{2,2}_{\mathrm{iso},j}(S;\mathbb{R}^3)$. Define $\tilde{y}(x) = y(A^{-1}x)$, where $A := \sqrt{(U_j^2)'}$. Clearly, $\tilde{y} \in W^{2,2}(AS;\mathbb{R}^3)$ and

$$\nabla \tilde{y}^T \nabla \tilde{y} = A^{-1} \nabla y^T \nabla y A^{-1} = \mathrm{Id}\,.$$

The set AS satisfies condition (2). Hence, by [17, Theorem 1] there exists a smooth isometric embedding $\tilde{\phi}$ for the flat metric Id such that $\|\tilde{y} - \tilde{\phi}\|_{W^{2,2}} \le \varepsilon \sqrt{\det(A)}$. Defining $\phi(x) := \tilde{\phi}(Ax)$ we get $\phi \in C^\infty(\bar{S};\mathbb{R}^3)$, $\nabla \phi^T \nabla \phi = A^2 = (U_j^2)'$, and $\|y - \phi\|_{W^{2,2}} \le \varepsilon$. A standard diagonal argument allows one to conclude. ■

Remark 4.1. Observe that the argument used in [16] to prove the existence of a recovery sequence cannot be applied here. Indeed, the truncation argument, which is the basis of the construction of [16], would lead to deformations with a low regularity, for which the penalty term cannot be written.

5 Convergence of minimizers with dead loads

In this section we prove Theorem 1.5 and Theorem 1.7. We start by showing that a sequence of deformations with bounded total energy has also bounded elastic energy. For the convenience of the reader, we recall that

$$\mathcal{M}_h = \operatorname*{argmax}_K F_h\,,$$

where

$$F_h(A) = \int_S f_h \cdot A \begin{pmatrix} x' \\ 0 \end{pmatrix} dx\,.$$

Similarly, $\mathcal{M}$ is defined as the set of maximizers over K of F, where

$$F(A) = \int_S f \cdot A \begin{pmatrix} x' \\ 0 \end{pmatrix} dx\,.$$

Lemma 5.1. *Let $\alpha \ge 2$ and $q > 1$. Suppose that $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ is a sequence of deformations that are quasi-minimizers for J_h^α, that is,*

$$\lim_{h\to 0} (J_h^\alpha(y_h) - \inf J_h^\alpha) = 0\,.$$

Then, $E_h^\alpha(y_h) \le C$ for every $h > 0$.

Proof. Firstly, we will prove that $h^\alpha E_h^\alpha(y_h) \to 0$. Fix $j \in \{1, \dots, l\}$ and let Q_h, R_h be, respectively, the rotation and the map given by in Proposition 3.4. Define

$$\tilde{y}_h = y_h - Q_h U_j \begin{pmatrix} x' \\ hx_3 \end{pmatrix} - c_h \,,$$

where

$$c_h = \frac{1}{|\Omega|} \int_\Omega \left(y_h - Q_h U_j \begin{pmatrix} x' \\ hx_3 \end{pmatrix} \right) dx \,.$$

Using the test deformation

$$(x', x_3) \mapsto U_j \begin{pmatrix} x' \\ hx_3 \end{pmatrix}$$

and the assumption (8), we get $\inf_y J_h^\alpha(y) \le Ch^{1-\gamma}$, so that

$$J_h^\alpha(y_h) - Ch^{1-\gamma} \le J_h^\alpha(y_h) - \inf_y J_h^\alpha(y) \le C \implies J_h^\alpha(y_h) \le Ch^{1-\gamma} \,.$$

We have

$$\begin{aligned} h^\alpha E_h^\alpha(y_h) &= h^\alpha J_h^\alpha(y_h) + \int_\Omega f_h \cdot \tilde{y}_h \, dx + \int_S f_h \cdot Q_h U_j \begin{pmatrix} x' \\ 0 \end{pmatrix} dx \\ &\le Ch^{\gamma+1} + Ch^{\gamma+1} \|\nabla_h \tilde{y}_h\|_{L^q} \,. \end{aligned} \tag{39}$$

Consider the set

$$\tilde{\Omega}_h = \{x \in \Omega \colon \operatorname{dist}(\nabla_h y_h, K) \ge 1\} \,.$$

By Proposition 3.4 and Remark 3.5, we deduce

$$\begin{aligned} \|\nabla_h \tilde{y}_h\|_{L^q}^q &= \int_\Omega |\nabla_h y_h - Q_h U_j|^q \, dx \le Ch^{-2} \int_\Omega \operatorname{dist}^q(\nabla_h y_h, K_j) \, dx \\ &\le Ch^{-2} \int_\Omega \operatorname{dist}^q(\nabla_h y_h, K) \, dx + Ch^{-2} \\ &\le Ch^{-2} \int_{\tilde{\Omega}_h} \operatorname{dist}^q(\nabla_h y_h, K) \, dx + Ch^{-2} \\ &\le Ch^{\alpha-2} E_h^\alpha(y_h) + Ch^{-2} \,, \end{aligned} \tag{40}$$

where we have used (W4). Combining (39)-(40) we get

$$h^\alpha E_h^\alpha(y_h) \le Ch^{\gamma+1} + Ch^{\gamma+1-\frac{2}{q}} (h^\alpha E_h^\alpha(y_h))^{\frac{1}{q}} + Ch^{\gamma+1-\frac{2}{q}} \,.$$

Recalling that $q > 1$, by Young's inequality we deduce that

$$h^\alpha E_h^\alpha(y_h) \le Ch^{\gamma+1} + Ch^{q'(\gamma+1-\frac{2}{q})} + Ch^{\gamma+1-\frac{2}{q}} \to 0 \,.$$

Hence, we can apply Proposition 3.1 and deduce that, at least along a subsequence, there is an index $j_0 \in \{1, \dots, l\}$ such that for $h \ll 1$

$$\int_\Omega \operatorname{dist}^2(\nabla_h y_h, K_{j_0}) \,, dx \le h^\alpha [E_h^\alpha(y_h) + (E_h^\alpha(y_h))^\theta] \,.$$

We now show that $E_h^\alpha(y_h) \le C$. To simplify the exposition, we suppose that $j_0 = j$. By Proposition 3.4 we have

$$\begin{aligned} \|\tilde{y}_h\|_{L^2}^2 &\le C \|\nabla \tilde{y}_h\|_{L^2}^2 \le Ch^{-2} \|\operatorname{dist}(\nabla_h y_h, K_j)\|_{L^2}^2 \\ &\le Ch^{\alpha-2} [E_h^\alpha(y_h) + (E_h^\alpha(y_h))^\theta] \,. \end{aligned} \tag{41}$$

Pick $\bar{R}_h U_{k_h} \in \mathcal{M}_h$ and define

$$\tilde{J}_h^\alpha(y) := E_h^\alpha(y) - \frac{1}{h^\alpha}\int_\Omega f_h \cdot y\,dx + \frac{1}{h^\alpha}\int_S f_h \cdot \bar{R}_h U_{k_h}\begin{pmatrix} x' \\ 0 \end{pmatrix} dx\,.$$

Note that y_h are quasi-minimizers for the functional $\tilde{J}_h^\alpha$. Moreover, using the test deformation

$$(x', x_3) \mapsto \bar{R}_h U_{k_h}\begin{pmatrix} x' \\ hx_3 \end{pmatrix},$$

we get $\inf_y \tilde{J}_h^\alpha(y) \le 0$, so that

$$\tilde{J}_h^\alpha(y_h) \le \tilde{J}_h^\alpha(y_h) - \inf_y \tilde{J}_h^\alpha(y) \le C\,.$$

Hence,

$$\begin{aligned} E_h^\alpha(y_h) &= \tilde{J}_h^\alpha(y_h) + \frac{1}{h^\alpha}\int_\Omega f_h \cdot y_h\,dx - \frac{1}{h^\alpha}\int_S f_h \cdot \bar{R}_h U_{k_h}\begin{pmatrix} x' \\ 0 \end{pmatrix} dx' \\ &= \tilde{J}_h^\alpha(y_h) + \frac{1}{h^\alpha}\int_\Omega f_h \cdot \tilde{y}_h\,dx + \frac{1}{h^\alpha}\int_S f_h \cdot (Q_h U_j - \bar{R}_h U_{k_h})\begin{pmatrix} x' \\ 0 \end{pmatrix} dx' \\ &\le C + Ch^{-\gamma-1}\|f_h\|_{L^2}(E_h^\alpha(y_h) + (E_h^\alpha(y_h))^\theta)^{\frac{1}{2}} \\ &\le C + C(E_h^\alpha(y_h))^{\frac{1}{2}} + C(E_h^\alpha(y_h))^{\frac{\theta}{2}}\,, \end{aligned}$$

where we have used (41) and the definition of $\mathcal{M}_h$. Since $0 < \frac{\theta}{2} < 1$, by an application of Young's inequality we conclude. ■

We are in a position to conclude the proof of Theorem 1.5.

Proof of Theorem 1.5. By Lemma 5.1 and Theorem 1.1–(i) there exists $j \in \{1, \dots, l\}$ and $y \in W^{2,2}(S;\mathbb{R}^3)$ satisfying $(\nabla y)^T \nabla y = (U_j^2)'$ such that, up to a subsequence, $(\nabla_h y_h) \to (\nabla y, \nu)$ in $L^2(\Omega;\mathbb{R}^{3\times 3})$, where ν is defined as in Lemma 3.6. Let

$$c_h = \frac{1}{|\Omega|}\int_\Omega y_h\,dx$$

and define $\tilde{y}_h := y_h - c_h$. By an application of the Poincaré-Wirtinger inequality, we deduce that $\tilde{y}_h \to y$ in $W^{1,2}(S;\mathbb{R}^3)$. By the strong convergence of $\frac{1}{h^2} f_h$, we get

$$\frac{1}{h^2}\int f_h \cdot y_h\,dx = \frac{1}{h^2}\int f_h \cdot \tilde{y}_h\,dx \to \int_S f \cdot y\,dx\,.$$

By Theorem 1.1–(ii) we immediately deduce that

$$\liminf_{h\to 0} J_h^2(y_h) \ge E_j^{\mathrm{K}}(y) - \int_S f \cdot y\,dx = J_j^{\mathrm{K}}(y)\,.$$

Now let $i \in \{1, \dots, l\}$ and $y' \in W^{2,2}(S;\mathbb{R}^3)$ be such that $(\nabla y')^T \nabla y' = (U_i^2)'$. Let $(\tilde{y}_h')$ be the recovery sequence for y' provided by Theorem 1.1–(iii). Then

$$\begin{aligned} J_j^{\mathrm{K}}(y) &\le \liminf_{h\to 0} J_h^2(y_h) \le \limsup_{h\to 0} J_h^2(y_h) = \limsup_{h\to 0}(\inf_y J_h^2(y)) \\ &\le \limsup_{h\to 0} J_h^2(y_h') = \lim_{h\to 0} J_h^2(y_h') = J_i^{\mathrm{K}}(y')\,, \end{aligned}$$

concluding the proof. ■